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FACIT-Fatigue Questionnaire Dataset Design and Endpoint Analysis in Clinical Trials

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ABSTRACT

Patient-Reported Outcomes (PROs) are essential in modern clinical research for capturing patients' health status directly from their perspective. The Functional Assessment of Chronic Illness Therapy–Fatigue (FACIT-Fatigue) Scale is a widely used 13-item PRO instrument that measures fatigue and its impact on daily activities over the past seven days, with total scores ranging from 0 to 52 (higher scores indicating less fatigue). This paper provides a step-by-step workflow for FACIT-Fatigue analysis in clinical trials. It describes the mapping of collected questionnaire responses to SDTM QS domain, the construction of the Analysis Data Model (ADaM) ADFACIT dataset, including reverse scoring, total score derivation with proration rules for missing items, and analysis visit assignment via time-based windowing and closest visit mapping methods. Finally, the paper outlines the analytical framework for evaluating fatigue endpoints, including descriptive statistics summarizing total scores by treatment group and time point, as well as a linear mixed-effects model incorporating treatment, time, and their interaction as fixed effects with baseline score as a covariate to estimate least squares mean differences between treatment groups.

INTRODUCTION

Patient-Reported Outcomes (PROs) are a critical component of modern healthcare and clinical research, capturing patients' health status directly from their perspective without interpretation by a clinician or anyone else. By using standardized questionnaires, PROs provide invaluable insights into the patient's experience, such as the true burden of a disease or the real-world impact of medical treatment on daily life. This direct feedback helps ensure that clinical trials measure outcomes that are meaningful to patients and supports regulatory agencies in evaluating the benefits and risks of new therapies. As a prominent example of a PRO measurement system, the Functional Assessment of Chronic Illness Therapy–Fatigue (FACIT-Fatigue) Scale[1] is a 13-item instrument designed to assess fatigue and its impact on daily activities and functioning over the past 7 days. As shown in Figure 1, it captures fatigue-related symptoms and their consequences, including tiredness, weakness, difficulty performing usual activities, lack of energy, and overall effects on quality of life.

The scale consists of Likert-type items scored from 0 to 4: "Not at all," "A little bit," "Somewhat," "Quite a bit," and "Very much." The total score is derived in ADaM and ranges from 0 to 52. Items 1, 2, 3, 4, 5, 6, 9, 10, 11, 12, and 13 are reverse scored which means a response of "4" becomes a "0", a "3" becomes a "1", and so on. Items 7 and 8 are scored directly. All item scores are then summed up to produce a total score between 0 and 52. A higher total score indicates a lower level of fatigue, whereas a lower score reflects greater fatigue.

This paper describes the step-by-step process of mapping collected FACIT-Fatigue data and incorporating them into the ADaM dataset to facilitate the endpoint analyses specified in the protocol.

FACIT Fatigue Scale

Below is a list of statements that other people with your illness have said are important. **Please circle or mark one number per line to indicate your response as it applies to the past 7 days.**

		Not at all	A little bit	Somewhat	Quite a bit	Very much
HI7	I feel fatigued	0	1	2	3	4
HI12	I feel weak all over	0	1	2	3	4
An1	I feel listless ("washed out")	0	1	2	3	4
An2	I feel tired	0	1	2	3	4
An3	I have trouble starting things because I am tired	0	1	2	3	4
An4	I have trouble finishing things because I am tired	0	1	2	3	4
An5	I have energy	0	1	2	3	4
An7	I am able to do my usual activities	0	1	2	3	4
An8	I need to sleep during the day	0	1	2	3	4
An12	I am too tired to eat	0	1	2	3	4
An14	I need help doing my usual activities	0	1	2	3	4
An15	I am frustrated by being too tired to do the things I want to do	0	1	2	3	4
An16	I have to limit my social activity because I am tired	0	1	2	3	4

Figure 1: FACIT Fatigue Scale (Version 4)

SDTM DATA MAPPED FROM THE QUESTIONNAIRE VALUES

The FACIT-Fatigue questionnaires are mapped to SDTM QS (Questionnaire) domain. Below is an example of the QS domain mapped from the collected values shown in Figure 2. CDISC SDTM Controlled Terminology should be used for the complete codelist definitions based on the SDTM IG versions. Each question is represented by QSTEST/QSTESTCD, while responses are captured in QSORRES (original result) and QSSTRESC/QSSTRESN (standardized results in character and numeric forms).

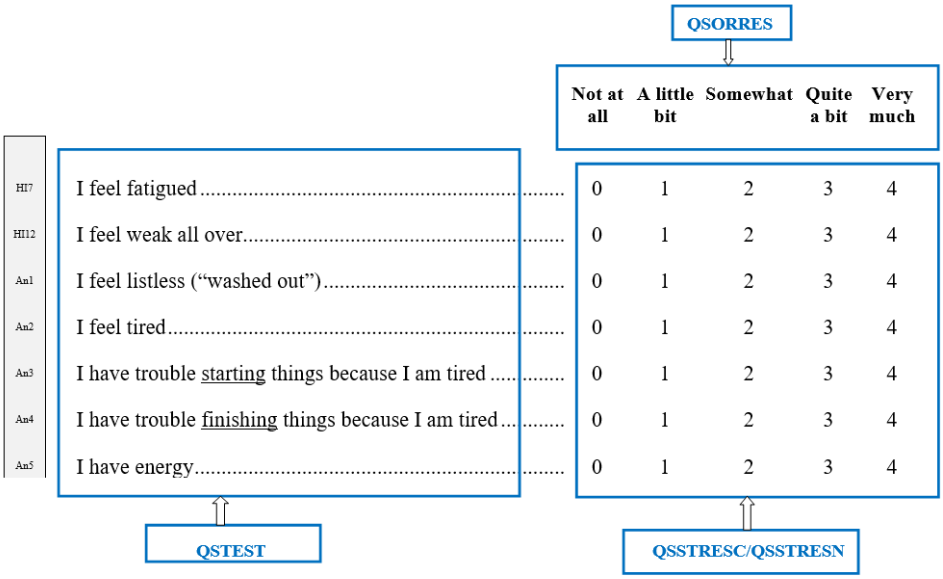


Figure 2: Example with FACIT-FATIGUE questions with SDTM annotation

USUBJID	DOMAIN	QSCAT	QSTEST	QSTESTCD	QSORRES	QSSTRESN
ABC-001	QS	FACIT-FATIGUE	I Feel Fatigued	FAC07001	Somewhat	2
ABC-001	QS	FACIT-FATIGUE	I Feel Weak All Over	FAC07002	Not at all	0
ABC-001	QS	FACIT-FATIGUE	I Have Energy	FAC07007	Somewhat	2
ABC-001	QS	FACIT-FATIGUE	Able to Do My Usual Activities	FAC07008	Quite a bit	3

Table 1: Example with QS domain mapped FACIT-FATIGUE values

ADAM WITH RECODED PARAMETERS

All the TLFs generated to support FACIT-Fatigue analysis will make use of ADaM dataset. To support this analysis, ADFACIT (FACIT-FATIGUE Analysis Dataset) is developed and belongs to the class of BDS (Basic Data Structure) and has a data structure of one record per subject per parameter per visit. All the questions that are mapped to QSTESTCD/QSTEST variables in SDTM/QS domain are directly mapped to PARAMCD/PARAM of ADFACIT. Along with directly mapped parameters, an additional "Total Score" parameter is derived to enable a more comprehensive assessment of fatigue and to support the protocol defined analysis of fatigue endpoints over the course of the trial. Rules on how to handle missing scores are defined by SAP. A common rule[2] for FACIT scales is: if less than 50% of the 13 items are answered then set "Total Score" to missing for that time point (e. g. if participant answers 6 or fewer items, then the "Total Score" is set to missing for that timepoint). If 7-12 items are answered, "Total Score" can be derived by (Sum of reversed score/number of answered items) X 13 (e.g. if 8 items are answered, then Total Score = (Sum of revered score/8) X13). If all 13 items are answered, Total Score is the sum of reverse-scored values.

PARAM	PARAMCD	PARAMN	Derivation for AVAL
I Feel Fatigued	FAC07001	1	If qstestcd='FAC07001' then AVAL=4-QS.QSSTRESN
I Feel Weak All Over	FAC07002	2	If qstestcd='FAC07002' then AVAL=4-QS.QSSTRESN
I Feel Listless	FAC07003	3	If qstestcd='FAC07003' then AVAL=4-QS.QSSTRESN
I Feel Tired	FAC07004	4	If qstestcd='FAC07004' then AVAL=4-QS.QSSTRESN
Trouble Starting Things	FAC07005	5	If qstestcd='FAC07005' then AVAL=4-QS.QSSTRESN
Trouble Finishing Things	FAC07006	6	If qstestcd='FAC07006' then AVAL=4-QS.QSSTRESN
I Have Energy	FAC07007	7	If qstestcd='FAC07007' then AVAL=QS.QSSTRESN
Able to Do My Usual Activities	FAC07008	8	If qstestcd='FAC07008' then AVAL=QS.QSSTRESN
I Need to Sleep During the Day	FAC07009	9	If qstestcd='FAC07009' then AVAL=4-QS.QSSTRESN
I Am Too Tired to Eat	FAC07010	10	If qstestcd='FAC07010' then AVAL=4-QS.QSSTRESN
Need Help Doing Usual Activities	FAC07011	11	If qstestcd='FAC07011' then AVAL=4-QS.QSSTRESN
Frustrated by Being Too Tired	FAC07012	12	If qstestcd='FAC07012' then AVAL=4-QS.QSSTRESN
Limit Social Activity, I am Tired	FAC07013	13	If qstestcd='FAC07013' then AVAL=4-QS.QSSTRESN
Total Score	FAC07014	14	AVAL=((sum of aval for PARAMN in (1 to 13) on the same date)/number of items that were answered) X 13 Note: if less than 7 individual PARAMCD have non-missing AVAL then AVAL is set to missing.

Table 2: PARAM PARAMCD and AVAL definitions

To analyze fatigue burden over the course of a clinical trial, FACIT-Fatigue data in the ADFACIT dataset needs to be mapped to standardized time points. There are two primary methods for deriving an analysis visit (AVISIT), Time-Based Windowing (using ADY) and Closest Visit Mapping (using Protocol-Defined Visits).

Using the time-based windowing method, ADY for each fatigue assessment is calculated. If ADY falls within a window defined by the SAP (e.g., Week 4 = Day 15 to Day 42), the assessment is assigned to the corresponding analysis visit. Sample code is shown below for the derivation of analysis visits based on ADY.

```
data qs_6;
set qs_5;
if aepoch="Extension"  then do;
ady4=adt-tr04sdt+(adt>=tr04sdt);
   if ady4=1 then do; avisit='Extension Week 0'; avisitn=1000; awtarget=1 ;  end;
   else if 57>=ady4>=2 then do; avisit='Extension Week 4'; avisitn=1004; awtarget=29 ; end;
   else if 127>=ady4>=58 then do; avisit='Extension Week 12'; avisitn=1012; awtarget=85 ; end;
   else if 211>=ady4>=128 then do; avisit='Extension Week 24'; avisitn=1024; awtarget=169 ; end;
   else if 302>=ady4>=212 then do; avisit='Extension Week 36'; avisitn=1036; awtarget=253 ; end;
   else if 358>=ady4>=303 then do; avisit='Extension Week 50'; avisitn=1050; awtarget=351 ; end;
   else if 407>=ady4>=359 then do; avisit='Extension Week 52'; avisitn=1052; awtarget=365 ; end;
end;
run;
```

The "Closest Visit Mapping" method derives an analysis visit by prioritizing the protocol's planned visit schedule as the primary anchor. It works by taking each fatigue assessment date and finding the protocol-defined visit (from the SV domain) that occurred closest in time to it. The assessment is then assigned the name and number of that closest planned visit, effectively linking the data point to the operational visit schedule rather than a standardized time window. In programming terms, each QS record is assigned with the VISIT name/number of the SV record that has the minimal time difference (e.g., abs(QSDTC − SVSTDTC)). Sample code is shown below for the derivation of analysis visits by “Closest Visit Mapping” method. In the sample code, a tie-breaking rule is also implemented. If multiple SV records with same time difference from QS record, then the SV record with “smaller” visit number will be used to assign VISIT name/number for QS records.

```
data qs_3;
merge qs_2( in=a )  dov1_t;
by usubjid;
if a;
run;

data qs_4_;
   set qs_3;

   array vdates{*} &visit_vars.;

   min_diff     = .;
   closest_date  = .;
   best_visitnum = .;
   length closest_visit $32;
   closest_visit = ' ';

   do i = 1 to dim(vdates);

      if vdates{i} = . then continue;

      diff = abs(adt - vdates{i});

      curr_visitnum = input(
                        scan(vname(vdates{i}), -1, '_'),
                        best.
                      );

      if min_diff = . then do;
         min_diff     = diff;
         closest_date  = vdates{i};
         closest_visit = vname(vdates{i});
         best_visitnum = curr_visitnum;
      end;
      else if diff < min_diff then do;
         min_diff     = diff;
         closest_date  = vdates{i};
         closest_visit = vname(vdates{i});
         best_visitnum = curr_visitnum;
      end;
      else if diff = min_diff and curr_visitnum < best_visitnum then do;
         closest_date  = vdates{i};
         closest_visit = vname(vdates{i});
         best_visitnum = curr_visitnum;
      end;

   end;

   format closest_date yymmdd10.;
   drop i diff ;
run;
```

ANALYSIS AND REPORTING

As one of the efficacy endpoints, the FACIT-Fatigue will be evaluated by assessing the change in total scores from baseline. A positive change in the FACIT-Fatigue score indicates improvement (i.e., reduced fatigue). One analytical approach is to summarize the "Total Score" using descriptive statistics and to apply a linear mixed-effects model to assess differences between treatment groups across multiple time points.

For descriptive statistics, FACIT-Fatigue "Total Score" was summarized by treatment group at each time point using mean, standard deviation, median, interquartile range, minimum, and maximum values. Change from baseline to each subsequent time point was also calculated for each treatment group along with corresponding summary statistics. Line plots were generated to visually display the trajectory of FACIT-Fatigue scores over time by treatment group.

For inferential analysis, a linear mixed-effects model was employed to evaluate differences in FACIT-Fatigue scores between treatment groups over time. The model included treatment group, time point, and their interaction (Treatment × Time) as fixed effects, with baseline FACIT-Fatigue score as a covariate. A random intercept for each subject was included to account for within-subject correlation due to repeated measurements. An unstructured covariance matrix was specified to model the within-subject correlation over time, and the Kenward-Roger method was used for denominator degrees of freedom estimation. Least squares (LS) mean differences between each active treatment group and placebo were estimated at each time point, along with corresponding 95% confidence intervals and p-values.

Above model is described as below in SAS code and sample display of report is also included below.

```
PROC MIXED DATA=adfacit;
  CLASS Subject Treatment Time;
  MODEL Change_from_Baseline = Treatment Time Treatment*Time Baseline / DDFM=KR;
  REPEATED Time / SUBJECT=Subject TYPE=UN;
  LSMEANS Treatment / PDIFF=CONTROL('Placebo') CL ALPHA=0.05;
RUN;
```

Treatment	N	Mean (SD) at Baseline	Mean (SD) at Week XX	Mean (SD) Change from Baseline at Week XX
Placebo	xx	xx.x (xx.x)	xx.x (xx.x)	xx.x (xx.x)
Treatment A	xx	xx.x (xx.x)	xx.x (xx.x)	xx.x (xx.x)
Treatment B	xx	xx.x (xx.x)	xx.x (xx.x)	xx.x (xx.x)
Treatment Comparison at Week XX			**LS Mean Difference (95% CI)**	***P-value***
Treatment A vs. Placebo			xx.x (xx.x, xx.x)	0.xxx
Treatment B vs. Placebo			xx.x (xx.x, xx.x)	0.xxx

CONCLUSION

In summary, this paper has presented an end-to-end workflow for handling FACIT-Fatigue data in clinical trials. By following this structured approach, statistical programmers and biostatisticians can ensure that fatigue endpoints are derived consistently, transparently, and in compliance with regulatory standards. The methods outlined here including time-based windowing and closest visit mapping for visit assignment, as well as the use of treatment-by-time interaction models with baseline adjustment provide a framework that could be adapted to other PRO instruments beyond the FACIT-Fatigue Scale. This workflow supports the generation of reliable, patient-centered evidence that informs treatment decisions and strengthens the overall quality of clinical trial submissions.

REFERENCE

1. Functional Assessment of Chronic Illness Therapy – Fatigue Scale (FACIT-Fatigue), Version 4, https://www.facit.org/measures/facit-fatigue
2. Scoring of the FACIT Measures, https://www.facit.org/scoring

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